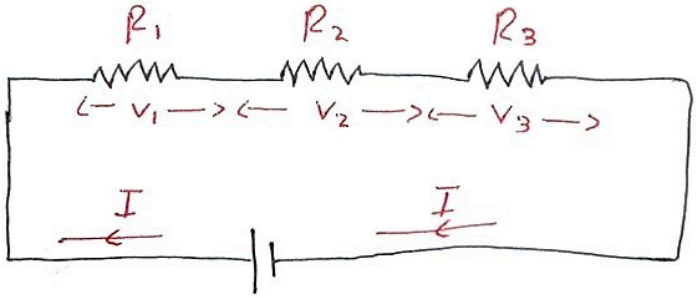
Current Electricity

⟹ Resistor Systems

→ Resistor systems in Series

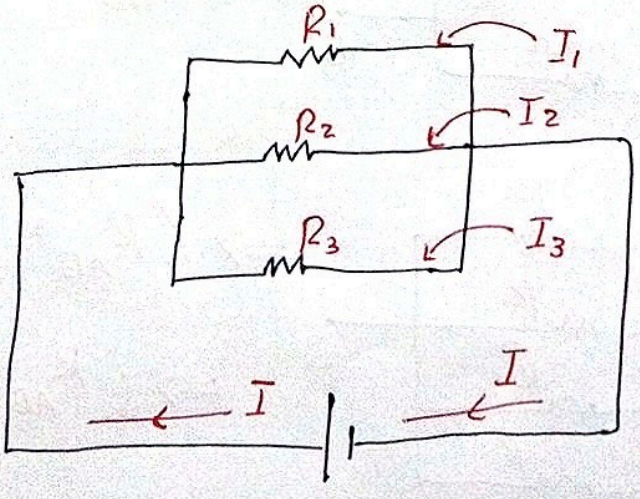


→ Same current goes through all resistors.

→ $V = V_1 + V_2 + V_3$

→ $R = R_1 + R_2 + R_3$

→ Resistor Systems in parallel



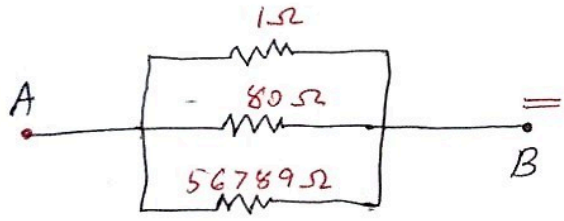
→ Some voltage goes through all resistors.

→ $$\boxed{I = I_1 + I_2 + I_3}$$

→ $$\boxed{\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

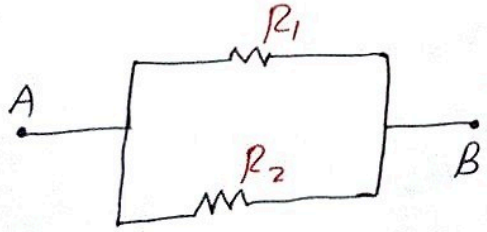
→ <u>Short Methods</u>.

(1)



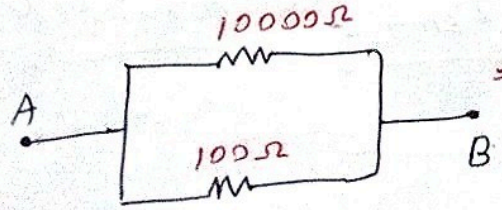
$\Rightarrow (R_{AB} < 1\Omega)$

(2)



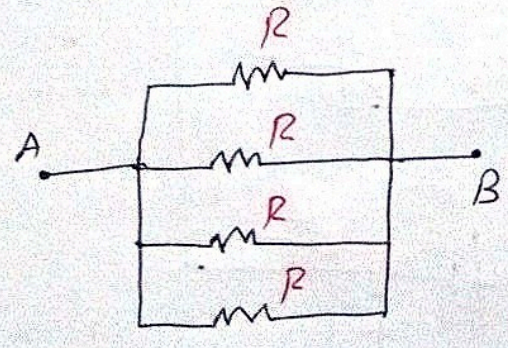
$\Rightarrow \left(R_{AB} = \frac{R_1 R_2}{R_1 + R_2}\right)$

(3)



$\Rightarrow (R_{AB} \approx 100\Omega)$

(4)



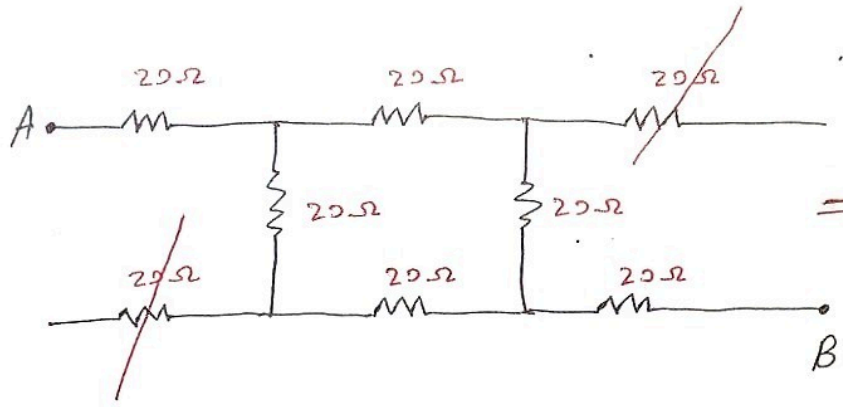
$\Rightarrow \left(R_{AB} = \frac{R}{4}\right)$

$\Rightarrow \left(R_{AB} = \frac{R}{n}\right)$

AIScanner

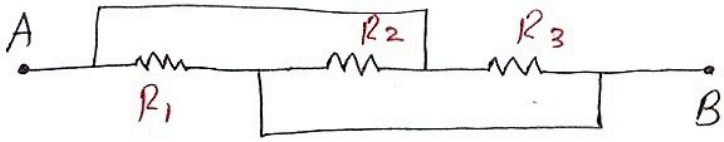
→ Important Methods

(1)

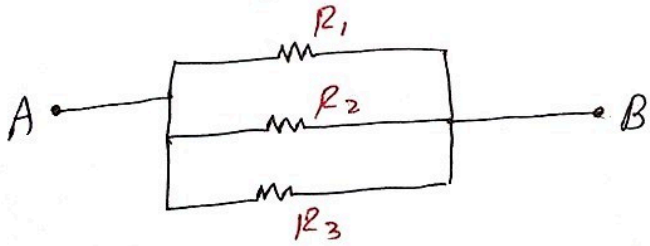


$\Rightarrow 60\Omega$

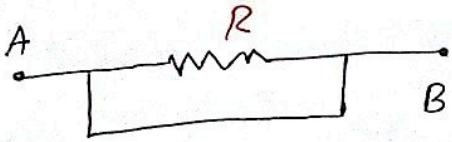
(2)



⇓

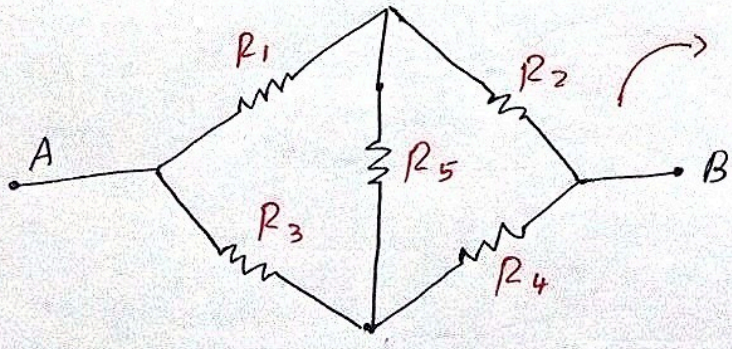


(3)



$\Rightarrow R_{AB} = 0\Omega$

(4) Wheastone bridge principle,



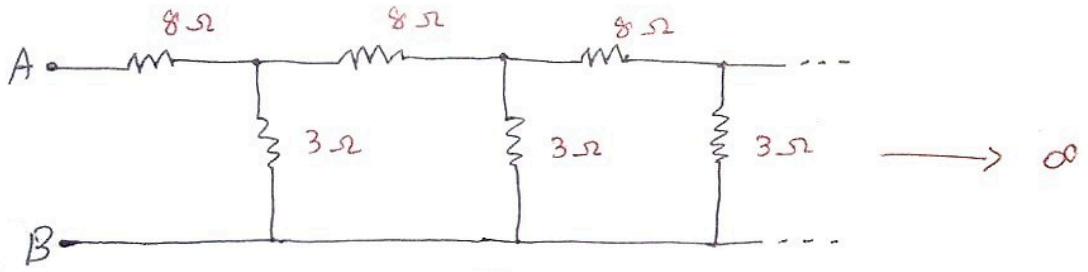
$$\frac{R_1}{R_2} = \frac{R_3}{R_4}$$

then,

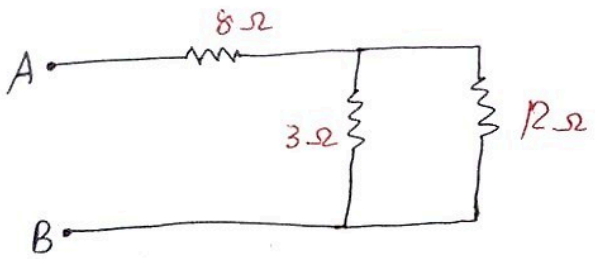
$$\boxed{V_C = V_D}$$

→ we can remove "R_5"

(5) When there are repeating Resistors,



⇓



$$\Rightarrow \left(R_{AB} = 4 + \left(\frac{3R}{3+R} \right) \right)$$